Bernardo Bahia Monteiro, PhD

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PhD in Aerospace Engineering from the University of Michigan. Dual citizenship: Portuguese and Brazilian

EXPERIENCE

Envision Energy — Boulder, CO, USA
Blade Aerodynamics, Loads, and Controls Engineer — July 2024–present

- Perform nonlinear aeroelastic simulations for load calculation of composite wind turbine blades (flexible multibody dynamics coupled with unsteady aerodynamics)
- Identify stability and resonance phenomena and propose structural changes to improve the blade's dynamics
- Test these changes in high-fidelity simulations
- Investigate performance issues that happen in the field
- Tool development (MATLAB/Python/Git) for Linux/Slurm HPC cluster (3000 cores)
  - Speed up loads post-processing by more than 6x, through code profiling and targeted optimization of critical code sections (reducing algorithmic complexity, etc.)
  - Propose new methods for loads analysis, stability identification, etc.
- Controller tuning

University of Michigan — Ann Arbor, MI, USA
Active Aeroelasticity and Structures Research Laboratory — September 2019–June 2024
Graduate Student Research Assistant funded by Airbus and NASA

- Developer of in-house nonlinear very-flexible aircraft simulation software (C++/Python) (simulation in UM/NAST considers 6DoF + flexible states + unsteady aerodynamics)
  - Selection of DAE solvers to guarantee fast calculations / numerical robustness / accuracy of the solution (e.g. adequate numerical damping)
  - Implementation of an algorithmic differentiation interface to allow use of simulation tool for design (MDO)
  - Implementation of linearization and closed-loop analysis in frequency domain
- Wind tunnel testing
  - Develop a real-time control system, integrating 3rd party sensors and actuators (C/C++)
  - Design, analysis, and operation of aero-servo-elastic wind tunnel model
  - Coordinate a 3 people design team
- Wing aerostructural design using MDO (Shell FEM + RANS CFD)
- Design and analysis of advanced subsonic and supersonic (very) flexible aircraft
- Develop novel control-related metrics for Multidisciplinary Design Optimization (MDO)
- Administrate the lab's Linux-based high-performance computers (3 HPCs)
- Write internal reports and present regularly to industry partners (Airbus and NASA)
- Write papers and present at international conferences

Embraer — São José dos Campos, SP, Brazil
Preliminary Design — February, 2019–June, 2019

- Program semi-empirical aerodynamic models for the in-house MDO code (Fortran)
- Attend design reviews

Flight Operations Engineering — January, 2018–January, 2019

- Calculate friction coefficient of a contaminated runway from flight data (Java)
- Identify unstable approaches from flight data, among other inflight events (Python)
- Support meetings with customers (airlines)

EDUCATION

University of Michigan — Ann Arbor, MI, USA
PhD Aerospace Engineering — September 2019–June 2024
Advisors: Carlos E. S. Cesnik, Ilya Kolmanovsky
Thesis: "Control Related Metrics for Multidisciplinary Design Optimization". PDF
MSE Aerospace Engineering — GPA: 4.0/4.0

Universidade Federal de Minas Gerais — Belo Horizonte, MG, Brazil

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BSc Aerospace Engineering — GPA: 4.9/5.0 February 2014–June 2019

Admitted in first place; graduated first in class

Thesis: "Simulation of Gas Turbine Engines". PDF

University of Strathclyde Glasgow, UK

One semester exchange in Aero-mechanical Engineering January–July, 2017

First class result in all disciplines, mostly at the master's level, including composites

Summer research assistant for the "Future UK Small Payload Launcher" project

PUBLICATIONS

[1] Bahia Monteiro, B., "Control Related Metrics for Multidisciplinary Design Optimization," Ph.D. thesis, University of Michigan, 2024. doi:10.7302/23748, PDF.

[2] Bahia Monteiro, B., "Fast Fatigue Load Evaluation for Wind Turbine Blades Considering Active Control," *1st Symposium on Vibration Fatigue and Related Topics*, CISM, Udine, Italy, 2025.

[3] Bahia Monteiro, B., Kolmonovsky, I., and Cesnik, C. E. S., "Controller Agnostic Design Metrics for Stochastic Disturbance Rejection," *AIAA SCITECH Forum*, 2024. doi:10.2514/6.2024-0901, PDF.

[4] Bahia Monteiro, B., Cesnik, C. E. S., and Kolmanovsky, I., "Gust Load Metrics for Multidisciplinary Design Optimization Considering Fatigue and Active Control," *RAeS 8th Aircraft Structural Design Conference*, 2023. PDF.

[5] Bahia Monteiro, B., Gray, A. C., Cesnik, C. E. S., Kolmanovsky, I., and Vetrano, F., "Bi-level Multidisciplinary Design Optimization of a Wing Considering Maneuver Load Alleviation and Flutter," *AIAA SCITECH 2023 Forum*, 2023, p. 0728. doi:10.2514/6.2023-0728, PDF.

[6] Bahia Monteiro, B., Cesnik, C., and Kolmanovsky, I., "Gust Load Alleviation Control Related Metric for Multidisciplinary Design Optimization," *IFASD*, 2022. PDF.

[7] Bahia Monteiro, B., "Simulation of Gas Turbine Engines," Bachelor's thesis, Universidade Federal de Minas Gerais, 2019. PDF.

[8] Jonsson, E., Riso, C., Monteiro, B. B., Gray, A. C., Martins, J. R. R. A., and Cesnik, C. E. S., "High-Fidelity Gradient-Based Wing Structural Optimization Including Geometrically Nonlinear Flutter Constraint," *AIAA Journal*, Vol. 61, No. 7, 2023, pp. 3045–3061. doi:10.2514/1.J061575, PDF.

CONTRIBUTIONS, AWARDS, AND DISTINCTIONS

- Reviewer for the AIAA Journal of Aircraft
- Reviewer for AIAA SciTech 2026 for 3 sessions:
  - Multidisciplinary Design Optimization: MBSE & MBSA
  - Dynamic Loads, Response, and Stability of Aerospace Vehicles
  - Structural Dynamics
- Gold medal award for best GPA of the class of 2019 at UFMG
- 2nd overall place on the Brazilian national high school exam, used for admission in university (ENEM 2013, more than 7 million students applied)

COMPUTER SKILLS

MATLAB/Simulink, Python, C/C++, Git, NASA's OpenMDAO, Linux/Unix (real-time), CI/CD, Fortran, Slurm, LaTeX

LANGUAGE SKILLS

English (fluent, C2 level), Portuguese (native), German (A2)

HOBBIES

Glider pilot (FAA PPG license), mountain biking, tennis, road cycling, skiing, rowing